

Main Tank Transformer Oil Standards (Natural Ester)						
Test Item	Standard	Method	Main Tank			
			Std. IEEE C57.106-2015	≤ 69 kV	> 69 kV - < 230 kV	≥ 230 kV
			Std. IEC 60422	≤ 72.5 kV	> 72.5 kV - ≤ 170 kV	> 170 kV - < 400 kV
II. New mineral oil in new equipment						
Dielectric breakdown	IEEE C57.106-2015	ASTM D1816	1 mm. gap (Spherically capped, kV)	≥ 25	≥ 30	≥ 32
	IEC 60422	IEC 60156	2 mm. gap (Spherically capped, kV)	≥ 45	≥ 52	≥ 55
			2.5 mm. gap (Spherical or Spherically capped, kV)	-	-	-
Dissipation factor (Power Factor)	IEEE C57.106-2015	ASTM D924	at 25 °C, % maximum	≤ 0.5		
	IEC 60422	IEC 60247	at 100 °C, % maximum	-		
			at 90 °C and 40-60 Hz			
Conductivity						
Water Content	IEEE C57.106-2015	ASTM D1533	ppm. (Mg/kg) maximum	≤ 300	≤ 150	≤ 100
	IEC 60422	IEC 60814	ppm. (Mg/kg) maximum	-		
Moisture in Oil						
Moisture in Paper						
Color	IEEE C57.106-2015	ASTM D1500	Units maximum	Clear		
	IEC 60422	ISO 2049	-			
Interfacial tension (IFT)	IEEE C57.106-2015	ASTM D971	mN/m	-		
	IEC 60422	ASTM D971, EN 14210	mN/m	-		
Neutralization number (Acidity)	IEEE C57.106-2015	ASTM D974	Mg KOH/g maximum	≤ 0.06		
	IEC 60422	IEC 62021-1 or IEC 62021-2	Mg KOH/g maximum	-		
Visual examination	IEEE C57.106-2015	ASTM D1524	-	Bright and Clear		
	IEC 60422	ISO 2049	-	Clear, free from sediment and suspended matter		
Oxidation inhibitor content	IEEE C57.106-2015	ASTM D2668	Type I oil, % maximum			
			Type II oil, % maximum			
Total dissolved gas	IEEE C57.106-2015	ASTM D3612	-			
Corrosive sulfur	IEEE C57.106-2015	ASTM D1275	-			
	IEC 60422	IEC 62535	-			
III. Continued use of in-service mineral oil						
Dielectric breakdown	IEEE C57.106-2015	ASTM D1816	1 mm. gap (Spherically capped), kV	≥ 23	≥ 28	≥ 30
	IEC 60422	IEC 60156	2 mm. gap (Spherically capped), kV	≥ 40	≥ 47	≥ 50
			2.5 mm. gap (Spherical or Spherically capped), kV	-	-	-
Dissipation factor (Power Factor)	IEEE C57.106-2015	ASTM D924	at 25 °C % maximum (%)	≤ 3		
			at 100 °C % maximum (%)	-		
Conductivity	IEC 61620	-	pS/m			
Water Content	IEEE C57.106-2015	ASTM D1533	ppm (Mg/kg) maximum	≤ 400	≤ 200	≤ 150
Moisture in Oil	IEEE 62-1995		%Saturation			
Moisture in Paper	IEEE 62-1995		%M/dw			
Color	IEEE C57.152-2013	ASTM D1500	-	≤ 2.5		
Interfacial tension (IFT)	IEEE C57.106-2015	ASTM D971	mN/m	-		
Neutralization number (Acidity)	IEEE C57.106-2015	ASTM D974	mg KOH/g maximum	≤ 0.3		
Corrosive sulfur		ASTM D1275	-			
Passivator	IEC 60666	-	ppm.			
Sludge	IEC60422	-	%w			
Furan (2-FAL)		ASTM D5837	ppb. (Open Type)			
			ppb. (Seal Type)			
DP	Chendong	-	-			
Oxidation inhibitor content	IEEE C57.106-2016	ASTM D2669	Type II mineral oil			
IV. Reconditioned or reclaimed mineral oil in transformers (After filling but before energizing)						
Dielectric breakdown	IEEE C57.106-2015	ASTM D1816	1 mm. gap (Spherically capped)	≥ 25	≥ 30	≥ 35
	IEC 60422	IEC 60156	2 mm. gap (Spherically capped)	≥ 45	≥ 55	≥ 60
			2.5 mm. gap (Spherical or Spherically capped)			
Water Content	IEEE C57.106-2015	ASTM D1533	ppm (Mg/kg) maximum	≤ 35	≤ 25	≤ 20
Interfacial tension (IFT)	IEEE C57.106-2015	ASTM D971	mN/m minimum	≥ 35		
Color	IEEE C57.106-2015	ASTM D1500	Maximum	≤ 1.5		
Visual examination	IEEE C57.106-2015	ASTM D1524	-	Clear		
Neutralization number (Acidity)	IEEE C57.106-2015	ASTM D974	mg KOH/g maximum	≤ 0.05		

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			Std. IEEE C57.106-2015			> 69 kV - < 230 kV			≥ 230 kV				
			Std. IEC 60422			≤ 72.5 kV			> 72.5 kV - ≤ 170 kV			> 170 kV - < 400 kV	
II. New mineral oil (Received in new equipment or after filling,prior to energization)				Normal	Monitor	Warning	Normal	Monitor	Warning	Normal	Monitor	Warning	
Dielectric breakdown	IEEE C57.106-2015	ASTM D1816	1 mm. gap (Spherically capped, kV)										
	IEC 60422	IEC 60156	2 mm. gap (Spherically capped, kV)										
Dissipation factor (Power Factor)	IEEE C57.106-2015	ASTM D924	at 25 °C, % maximum										
	IEC 60422	IEC 60247	at 100 °C, % maximum										
Conductivity			at 90 °C and 40-60 Hz										
Water Content	IEEE C57.106-2015	ASTM D1533	ppm. (Mg/kg) maximum										
Moisture in Oil	IEC 60422	IEC 60814	ppm. (Mg/kg) maximum										
Moisture in Paper													
Color	IEEE C57.106-2015	ASTM D1500	Units maximum										
	IEC 60422	ISO 2049	-										
Interfacial tension (IFT)	IEEE C57.106-2015	ASTM D971	mN/m										
	IEC 60422	ASTM D971, EN 14210	mN/m										
Neutralization number (Acidity)	IEEE C57.106-2015	ASTM D974	Mg KOH/g maximum										
	IEC 60422	IEC 62021-1 or IEC 62021-2	Mg KOH/g maximum										
Visual examination	IEEE C57.106-2015	ASTM D1524	-										
	IEC 60422	ISO 2049	-										
Oxidation inhibitor content	IEEE C57.106-2015	ASTM D2668	Type I oil, % maximum										
			Type II oil, % maximum										
Total dissolved gas	IEEE C57.106-2015	ASTM D3612	-										
Corrosive sulfur	IEEE C57.106-2015	ASTM D1275	-										
	IEC 60422	IEC 62535	-										
III. Continued use of in-service mineral oil													
Dielectric breakdown	IEEE C57.106-2015	ASTM D1816	1 mm. gap (Spherically capped), kV	28	28-23	23	33	33-28	28	35	35-30	30	
			2 mm. gap (Spherically capped), kV	45	45-40	40	52	52-47	47	55	55-50	50	
	IEC 60422	IEC 60156	2.5 mm. gap (Spherical or Spherically capped), kV	-	-	-	-	-	-	-	-	-	
Dissipation factor (Power Factor)	IEEE C57.106-2015	ASTM D924	at 25 °C % maximum (%)	2	2-3	3	2	2-3	3	2	2-3	3	
			at 100 °C % maximum (%)	-	---	-	-	---	-	-	---	-	
Conductivity	IEC 61620	-	pS/m	-	-	-	-	-	-	-	-	-	
Water Content	IEEE C57.106-2015	ASTM D1533	ppm (Mg/kg) maximum	300	300-400	400	150	150-200	200	100	100-150	150	
Moisture in Oil	IEEE 62-1995		%Sataration	-	-	-	-	-	-	-	-	-	
Moisture in Paper	IEEE 62-1995		%M/dw	-	-	-	-	-	-	-	-	-	
Color	IEEE C57.152-2013	ASTM D1500	-	2	2-2.5	2.5	2	2-2.5	2.5	2	2-2.5	2.5	
Interfacial tension (IFT)	IEEE C57.106-2015	ASTM D971	mN/m	-	-	-	-	-	-	-	-	-	
Neutralization number (Acidity)	IEEE C57.106-2015	ASTM D974	mg KOH/g maximum	0.2	0.2-0.3	0.3	0.2	0.2-0.3	0.3	0.2	0.2-0.3	0.3	
Corrosive sulfur		ASTM D1275	-	-	-	-	-	-	-	-	-	-	
Passivator	IEC 60666		ppm.	-	-	-	-	-	-	-	-	-	
Sludge	IEC60422	-	%w	-	-	-	-	-	-	-	-	-	
Furan (2-FAL)		ASTM D5837	ppb. (Open Type)	-	-	-	-	-	-	-	-	-	
			ppb. (Seal Type)	-	-	-	-	-	-	-	-	-	
DP	Chendong	-	-	-	-	-	-	-	-	-	-	-	
Oxidation inhibitor content	IEEE C57.106-2016	ASTM D2669	Type II mineral oil	-	-	-	-	-	-	-	-	-	
IV. Reconditioned or reclaimed mineral oil in transformers (After filling but before energizing)													
Dielectric breakdown	IEEE C57.106-2015	ASTM D1816	1 mm. gap (Spherically capped)		-			-			-		
			2 mm. gap (Spherically capped)		-			-			-		
	IEC 60422	IEC 60156	2.5 mm. gap (Spherical or Spherically capped)		-			-			-		
Water Content	IEEE C57.106-2015	ASTM D1533	ppm (Mg/kg) maximum		-			-			-		
Interfacial tension (IFT)	IEEE C57.106-2015	ASTM D971	mN/m minimum		-			-			-		
Color	IEEE C57.106-2015	ASTM D1500	Maximum		-			-			-		
Visual examination	IEEE C57.106-2015	ASTM D1524	-	Clear									
Neutralization number (Acidity)	IEEE C57.106-2015	ASTM D974	mg KOH/g maximum		-			-			-		